

Koketso Sethosa

Exercise 3 : SQL Case Statement

```
SELECT product_name ,  
       price ,  
       CASE  
         WHEN price > 1000 THEN 'Expensive'  
         WHEN price  
           BETWEEN 100 AND 1000  
         ELSE THEN 'Mid-range'  
         ELSE 'Budget'  
       END AS price_category  
FROM products ;
```

Output Table

product_name	price	price - category
Laptop	1200,00	Expensive
Phone	800,00	Mid-range
Keyboard	45,00	Budget
Monitor	300,00	Mid-range
Mouse	25,00	Budget.

```

SELECT customer_name,
       amount,
       CASE
         WHEN amount ≥ 1000 THEN 'High Value'
         WHEN amount
           BETWEEN 500 AND 999,99
           THEN 'Medium Value'
         Else 'Low Value'
       END AS order_value_category
FROM orders ;

```

Output Tax

Customer		
customer_name	amount	order_value_category
Alice	150,00	Low Value
Bob	560,00	Medium Value
Charlie	999,99	Medium Value
Diana	45,50	Low Value
Ethan	1200,00	High Value

```

3) SELECT emp-name,
           department,
           salary,
           CASE
             WHEN department = 'IT'
               AND salary > 80 000
               THEN 'Senior IT'
             WHEN department = 'Experienced HR'
               AND salary > 55 000
               THEN 'Experienced HR'
             ELSE 'Staff'
           END AS position-level
FROM HR employees;

```

Output Table

emp-name	department	salary	position-level
John	IT	85 000	Senior IT
Sara	HR	60 000	Experienced HR
Mark	IT	75 000	Staff
Lucy	Finance	95 000	Staff
Tom	HR	55 000	Staff

```

4) SELECT student_name,
       score,
       CASE
         WHEN score  $\geq$  90 THEN 'A'
         WHEN score BETWEEN 80 AND 89 THEN 'B'
         WHEN score BETWEEN 70 AND 79 THEN 'C'
         WHEN score BETWEEN 60 AND 69 THEN 'D'
         ELSE 'F'
       END AS grade
FROM students;

```

Output Table

student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

```

Q5) SELECT delivery-id,
        delivery-time-minutes AS dtm,
        CASE
            WHEN dtm <= 30 THEN 'Fast'
            WHEN dtm BETWEEN 31 AND 60 THEN 'On Time'
            ELSE 'Late'
        END AS performance
FROM deliveries;

```

Output Table

delivery-id	delivery ^{dtm}	performance
1	45	On Time
2	80	Late
3	30	FAST
4	65	Late
5	100	Late.

```

SELECT issue_type,
       priority,
       CASE
         WHEN priority = 1 THEN 'Low'
         WHEN priority = 2 THEN 'Medium'
         ELSE 'High'
       END AS priority_label
FROM tickets

```

Output Table

issue_type	priority	priority_label
1 login issue	1	Low
2 Server down	3	High
3 Slow system	2	Medium
4 Email error	2	Medium
5 Password reset	1	Low

```

7) SELECT student_id,
        100 * ( days - present / total - days ) AS attendance - percentage,
        CASE
            WHEN attendance - percentage >= 90
            THEN 'Excellent'
            WHEN attendance - percentage
                BETWEEN 75 AND 89
            THEN 'Good'
            ELSE 'Needs Improvements'
        END AS attendance - status
FROM attendance

```

Output Table :

student_id	attendance - percentage	attendance - status
1	90	Excellent
2	60	Good
3	96	Excellent
4	50	Needs Improvement
5	100	Excellent.

8) SELECT

product-id,
stock-qty,
CASE

~~WHEN stock-qty = 0 THEN~~

WHEN stock-qty = 0 THEN 'Out of Stock'

WHEN stock-qty

BETWEEN 1 AND 5 ~~THEN~~

THEN 'Low Stock'

ELSE 'In Stock'

END AS stock-status

FROM products-inventory;

product-id	stock-qty	stock-status
1	5	Low Stock
2	0	Out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

```

9) SELECT class subject,
        enrolled_students AS e_s,
        CASE
            WHEN e_s >= 25 THEN 'Large'
            WHEN e_s BETWEEN 10 AND 24
                THEN 'Medium'
            ELSE 'Small'
        END AS class_size_category
FROM classes;

```

Output Table

subject	enrolled-st e-s	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

```

10) SELECT payment_id,
           payment_method,
           amount,
           CASE
             WHEN payment_method = 'Cash'
                AND amount >= 200
             THEN 'Eligible for Discount'
             ELSE 'Not Eligible'
           END AS discount_eligibility
FROM payments;

```

Output Table

payment_id	payment_method	amount	discount_eligibility
1	Card	50,00	Not Eligible
2	Cash	200,00	Eligible for Discount
3	Card	150,00	Not Eligible
4	PayPal	75,00	Not Eligible.
5	Cash	300,00	Eligible for Discount